

Predictive Insight Engine

1 What are Supervised Learning Algorithms?

Supervised learning algorithms learn a mapping from input features (X) to a known output or label (y) using a labeled training dataset. The model is trained by comparing its predictions against the true labels and adjusting itself to reduce that error, so it can later predict outputs for new, unseen inputs.

2 Difference between Regression Algorithms vs Classification Algorithms.

Regression algorithms predict a continuous numeric value, such as a house price or a temperature. Classification algorithms predict a discrete category or class label, such as spam vs. not spam. The core difference lies in the type of target variable being predicted: continuous for regression, categorical for classification.

3 Explain Simple Linear Regression.

Simple Linear Regression models the relationship between one independent variable (X) and one dependent variable (y) as a straight line: $y = mX + c$, where m is the slope (the change in y per unit change in X) and c is the intercept (the value of y when X = 0). The line is fitted by minimizing the sum of squared errors between the predicted and actual values of y.

4 List and explain the Assumptions of Linear Regression.

- **Linearity** — the relationship between the independent and dependent variables is linear.
- **Independence of errors** — residuals are not correlated with one another.
- **Homoscedasticity** — residuals have roughly constant variance across all fitted values.
- **Normality of residuals** — the residuals are approximately normally distributed.
- **No or low multicollinearity** — independent variables are not highly correlated with each other, relevant when using multiple predictors.

5 What is Bias-Variance Trade-Off?

Bias is the error introduced by overly simplistic assumptions in a model, causing it to underfit and miss real patterns in the data. Variance is the error introduced by a model being too sensitive to the training data, causing it to overfit and capture noise instead of signal. Since total error is roughly the sum of bias squared, variance, and irreducible error, reducing one often increases the other — the goal is to find a model complexity that balances both.

6 Explain Overfitting and Underfitting with examples.

Overfitting occurs when a model learns the training data too closely, including its noise, so it performs very well on training data but poorly on unseen test data. Example: a very high-degree polynomial regression fit to house prices that curves to follow every fluctuation in the training set.

Underfitting occurs when a model is too simple to capture the underlying pattern in the data, so it performs poorly on both training and test data. Example: fitting a simple linear regression on a single weak feature when the true price actually depends on many factors.